
Design of a Comprehensive Web-Based Academic Management Platform with Integrated Intelligent Features as an Alternative for Traditional LMS

DIPLOMA THESIS

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Technical Overview

Context and motivation. The diploma project resulted in the development of **UniLMS** — a comprehensive web-based academic platform built as a modern alternative to traditional learning management systems such as Moodle, Canvas, and Blackboard. An empirical survey of 326 stakeholders in Kazakhstan (209 students and 117 teachers) confirmed three pain points: 72.7 % of respondents miss deadlines in their current systems, 59.2 % want a single unified platform, and 69.9 % are willing to adopt a university-specific LMS. These findings, together with a competitive analysis of seven existing platforms, defined three guiding principles: a contemporary user experience, native artificial-intelligence integration in core academic workflows, and complete trilingual coverage in English, Russian, and Kazakh.

Architecture. The system is delivered as a modular monolith with three clearly separated layers and ships as a three-service Docker Compose stack. The presentation layer is built with **Next.js 14** (App Router), **Tailwind CSS 3.4**, **TanStack Query 5** for server-state management, and **Framer Motion** for transitions. A custom design system based on the OKLCH color space provides perceptually uniform gradients, dual themes (forest light, violet dark), a density toggle, and a typography scale built on Geist and Instrument Serif. The application layer is implemented in **NestJS 10** with modular architecture, dependency injection, DTO validation through class-validator and Zod, role-based access control for student, teacher, and administrator roles, and Server-Sent Events for sub-second real-time notifications. **PostgreSQL 15** with **Prisma 5.22** forms the data layer, with fourteen domain models, ACID transactions, full-text search, and automatic migrations on container startup.

AI module. The intelligent layer is powered by **Anthropic Claude** and exposes five production endpoints: structured assignment feedback with strengths and improvement areas; topic-based quiz generation; course summarization with workload estimation; individual student performance analysis with low / medium / high risk scoring; and a streaming AI chat assistant that delivers responses token by token through SSE and supports client-side cancellation via **AbortController**. Every endpoint accepts a language parameter, and a built-in demo mode allows the platform to run without any external API key.

Security, localization, and deployment. Authentication relies on JWT access and refresh tokens stored in httpOnly cookies, with bcrypt hashing, refresh-token rotation, Helmet headers, and a Throttler rate limit of one hundred requests per minute. Every administrative action is recorded in an immutable audit log. Localization covers UI strings, AI prompts and replies, e-mails, and date / number formatting through the **Intl** API with the **kk-KZ** locale. End-to-end tests use Jest and Supertest, the project compiles in strict TypeScript mode with zero errors, and the entire stack launches with a single `docker compose up` command.

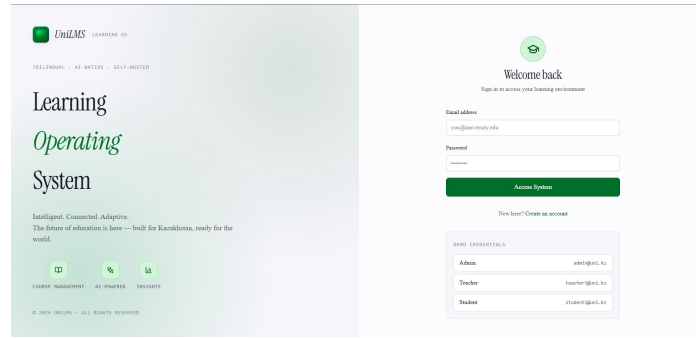


Figure 1: Authentication screen with email/password login

The login screen is the entry point of the platform and demonstrates the custom design system in action: an Instrument Serif heading, a clean form built from primitives of the design system, and an inline error-handling pattern shared across the application. Authentication itself is performed against the NestJS backend using a JWT pair stored in httpOnly cookies, with bcrypt password verification and refresh-token rotation on every renewal.

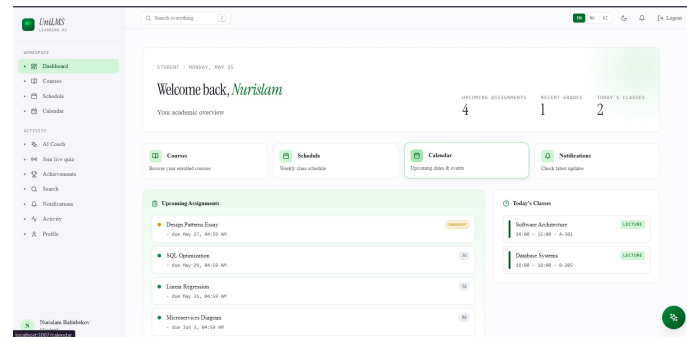


Figure 2: Student dashboard with active courses, deadlines, and recent activity

The student dashboard aggregates the most important information for a learner in a single view: enrolled courses with progress indicators, upcoming assignment deadlines, recent grades, and unread notifications. Every widget is rendered with optimistic UI through TanStack Query so that interactions remain instantaneous even on a slow connection.

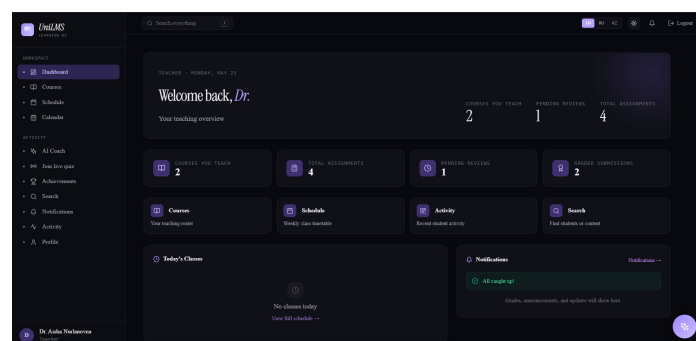


Figure 3: Teacher dashboard with course overview and pending submissions

The teacher dashboard mirrors the student view but is oriented toward instruction: it surfaces the number of pending submissions to grade, average class performance per course, and quick actions for creating new assignments or quizzes. Role-based access control enforced on the NestJS side guarantees that only users with the teacher role can reach this route.

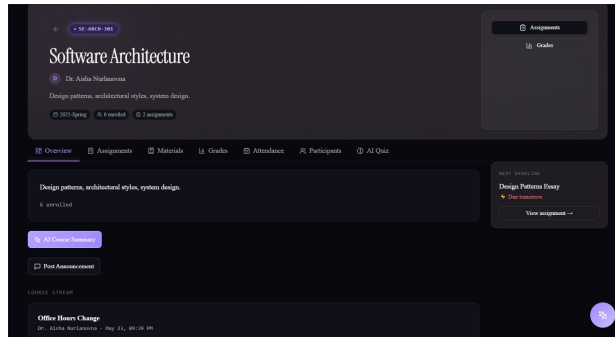


Figure 4: Course detail page with materials, modules, and progress tracking

The course detail page is the central learning hub. It presents a syllabus broken down into modules, downloadable materials, embedded videos, and a progress tracker that updates as the student completes activities. The same page acts as the teacher’s content-management interface: editable in place under the teacher role, read-only under the student role.

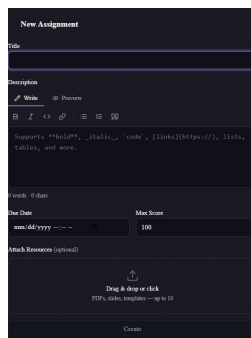


Figure 5: Assignment creation interface from the teacher perspective

When a teacher creates a new assignment, the form collects the title, description, attachments, allowed file types, due date, and grading scheme. All values are validated on both the client (Zod schema) and the server (class-validator DTO) so that invalid data can never reach the database. Created assignments immediately trigger SSE notifications to enrolled students.

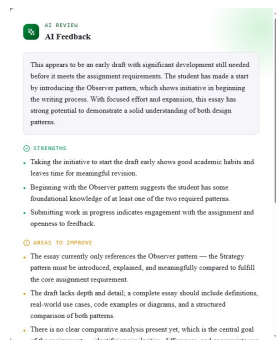


Figure 6: AI-generated structured feedback on a student submission

The first of five AI endpoints generates structured feedback for a student submission. The model returns an overall assessment, a list of strengths, a list of areas that require improvement, and concrete actionable suggestions. The output is validated at runtime through a Zod schema, which guarantees that downstream UI code always receives well-formed data.

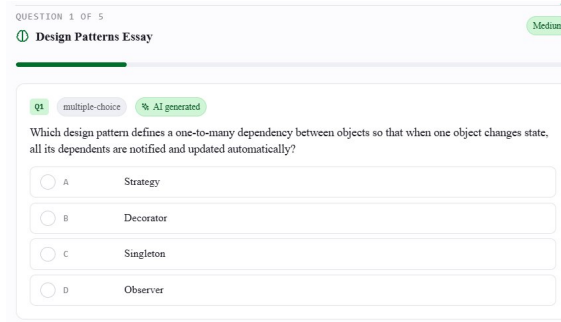


Figure 7: AI quiz generation with topic, count, difficulty, and language

The quiz generator accepts a topic, the desired number of questions, a difficulty level, and a target language (EN, RU, or KZ). Anthropic Claude synthesizes a quiz with multiple-choice questions, correct answers, and short explanations. The result can be assigned to a course with one click, which turns the AI-generated content into a regular interactive activity.

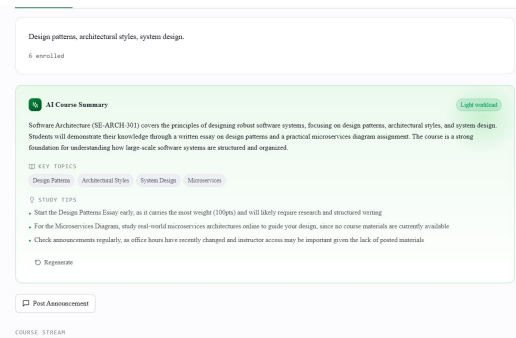


Figure 8: AI-generated course summary with key topics and workload estimation

Course summarization compresses an entire course into a synthesized overview containing key topics, study tips, and a workload estimation classified as light, moderate, or heavy. Teachers use this view to refine their curriculum; students use it for last-minute revision or to decide whether to enrol.

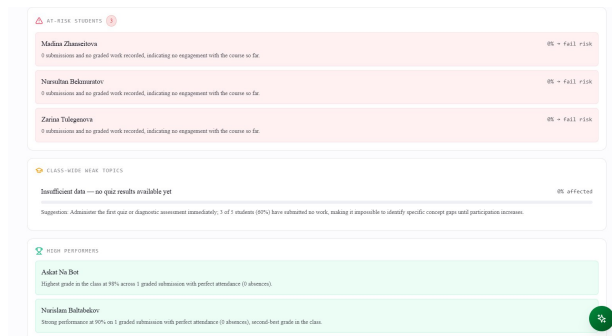


Figure 9: AI Student Analysis screen with risk level and recommendations

Student Analysis combines grades, submission timeliness, and participation metrics, then asks the model to classify the learner's risk level as low, medium, or high and to produce a list of personalized recommendations. This feature allows teachers to intervene early with students who would otherwise silently drift off track.

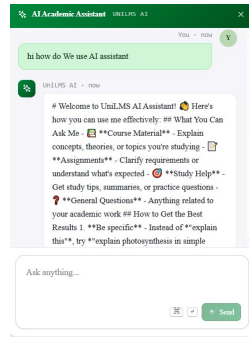


Figure 10: Streaming AI chat assistant with Server-Sent Events

The streaming AI chat is the most interactive feature: it delivers responses token by token through Server-Sent Events, supports mid-stream cancellation through `AbortController`, and keeps the conversation history in PostgreSQL so that students can resume previous sessions. The model automatically replies in the language of the user's last message.

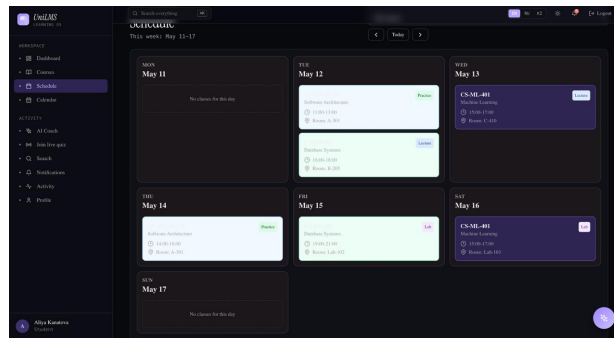


Figure 11: Weekly schedule and calendar with class, deadline, and event overlays

The schedule module unifies recurring class slots, one-off events, assignment deadlines, and quiz windows on a single weekly grid. Dates and times are formatted through the Intl API with the user's preferred locale, including correct Kazakh weekday names and date abbreviations.

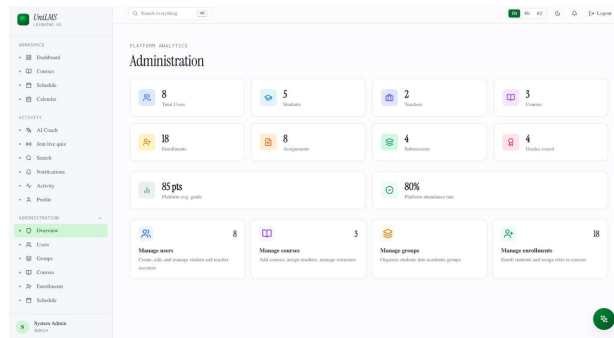


Figure 12: Administrative dashboard with platform-wide statistics

The administrative dashboard aggregates platform-wide statistics — total users by role, active courses, recent registrations, and AI-endpoint usage. Every administrative action is recorded in an immutable audit log, which is critical for accountability in a real university deployment.